CostVision
AI COST INTELLIGENCE

CostVision Implementation Blueprint

Secure enterprise deployment inside our network — and integration with CAPEE.
All CAD models, drawings and images stay inside the company. Verified in the code.

100% CAD stays internal

AI controls BUILT & tested

~3-5 weeks remaining

6-phase rollout

Management briefing · July 2026 · Grounded in a line-by-line audit of the platform code

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PURPOSE OF THIS MEETING

The decision we are asking for today

Approve Phases 1–2: the IT-Security assessment and the architecture design for deploying CostVision inside our network and connecting it to CAPEE. (~5–7 weeks, existing teams, no licence spend.)

Why now

The tool is built, tested (1,005 automated tests) and proven on live runs — the value is waiting on deployment, not development.

Why it is safe

CAD processing already runs fully inside the server. The private AI routing and the air-gapped switch are now BUILT and tested live.

Why CAPEE wins

CAPEE keeps the workflow; CostVision adds the AI, physics and organisational memory behind it. No tool replacement.

BACKGROUND

What CostVision is — a quick recap

18 cost engines

Machining, casting, injection, PCB, software and more — physics-based, every figure traceable

CAD-to-Cost

Reads STEP/IGES models with a real geometry engine and auto-fills the costing

Photo-to-BOM

Costs a PCB from photographs — detects components and builds the bill of materials

Agentic AI (upgraded 2026)

Learns from every quote; conformal confidence with a guarantee, a causal negotiation coach, outcome-weighted findings and what-if scenarios — all glass-box

Proven, not promised: estimating error cut from 10.9% to 0.3% after learning from 3 real quotes · £512k/yr of pricing issues found autonomously in the live demo · 1,005 automated tests protect it all.

THE CONCEPT · EXPLAINED

Agentic vs autonomous agentic — with examples

Two levels of "agentic". On the left the AI acts because you asked and you stay in the loop; on the right it acts on its own, unattended, within limits you set.

Agentic AI

Assisted — you trigger it and stay in the loop; a human approves

AI Agent Describe a part in plain English — it builds the full cost model for you.

CAD / photo → cost Upload a STEP file — it measures the geometry and interprets material & process.

Negotiation coach Open a part — it drafts the buyer's counter-argument and target price.

Rate-data assistant Ask a costing question — it answers from your own rates, with citations.

Autonomous Agentic AI

Self-directed — runs unattended on a schedule, within limits you set

Savings monitor Runs on the server, compares paid vs should-cost, opens findings itself — £0.5M/yr found unattended.

Self-audit Re-checks every estimate for known errors and corrects within bounds — nobody asks it to.

Calibration & drift Learns from logged quotes, re-derives factors, and watches for drift continuously.

Outcome-weighted ranking Learns which findings actually convert and re-prioritises the queue on its own.

The common thread: you set the boundaries and every action stays glass-box and auditable — autonomy never means the AI sets a price in secret.

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BACKGROUND · NEW IN 2026

Latest intelligence — six upgrades that widen the moat

Self-audit & guardrails	A deterministic layer re-checks every estimate for known errors and applies bounded corrections — proven on five real automotive CAD parts (a fuel tank fell from £217 to £25). Geometry stays ground truth; the AI never overrules it.
Learns from your quotes	Bulk-import logged actuals; the engine builds per-segment calibration and watches for drift, so the number tightens as your own history grows — and status rides on every estimate.
Negotiation coach	Names the cost driver and writes the counter: "a quote of £95 implies aluminium +14% above spot — hold at £86.34."
Outcome-weighted findings	The agent learns which findings convert and ranks by money it can really recover — a £100k gap that closes beats a £200k gap that never does.
What-if engine	Drag a commodity ±20% and the price recomputes live; run it across the whole portfolio ("if steel +10%, 7 parts go underwater").
Glass-box by design	Every learned or derived number stays auditable — no black-box weight ever touches the price. The competitor edge is the opposite.

TECHNOLOGY, SIMPLY EXPLAINED

What the tool is made of — in plain words

Frontend *What users see*	The web application in the browser — forms, dashboards, reports. Nothing is installed on laptops; it is served from our own server.
Backend *The engine room*	The server application that does the work: runs the 18 cost engines, the AI logic and all calculations. Runs on a standard company virtual machine.
Database *The memory*	Where rate libraries, saved costings and the AI knowledge base live. Standard corporate database (PostgreSQL), encrypted, backed up by IT.
CAD engine *The 3D model reader*	Specialist geometry software (OCCT — the same core used by CAD vendors) that measures the 3D model: volume, weight, walls, holes. Runs INSIDE the backend.
AI layer *The language brain*	The AI model used for vision and language tasks. NOW BUILT: one switchboard controls every AI call — route it to a private endpoint, or turn it fully OFF (air-gapped).

SECURITY

The security requirement — and the key finding

IT-Security requirement: **no CAD files, engineering drawings or images may leave the company network. Ever.**

The key finding from the code audit: **CAD files already never leave the server.**

The geometry engine runs inside our backend and processes CAD models in memory — files are not even written to disk, let alone sent anywhere. The only outbound flows are the AI layer (photos and derived summaries) and two optional public feeds. The controls that close this gap are now BUILT: private AI routing and a provable air-gapped switch, tested live in all three modes.

Why you can trust this: this is not a vendor claim — we audited every outbound network call in the platform's source code, line by line, and listed each one. The full inventory is in the written plan.

DATA-FLOW MAP

Where data flows today — verified in the code

STAYS INSIDE — already ✓

CAD / geometry processing
OCCT engine runs in the backend; files processed in memory, never stored

All 18 cost engines
Physics + maths, fully local

Knowledge base & learning loop
Similarity, calibration, autonomous findings — local database

BOM file parsing, exports, reports
Local parsers, local PDF/Excel generation

LEAVES TODAY — controlled ✓

AI layer calls
Photos + CAD-derived summaries. Private routing BUILT — one setting points every call at our endpoint

Live part pricing (optional, OFF by default)
Part-number text only — silenced by the air-gap switch

News ticker feeds
Public news, no company data — silenced by the air-gap switch

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DECISION

Deployment options — our recommendation

OPTION A

Fully air-gapped

Everything on our VMs. AI features off via the built-in AIR-GAPPED switch (already implemented and tested). Deterministic engines fully working.

+ Zero external connections — provable today, switch is built

- Reduced AI capability

OPTION B ★ RECOMMENDED

Private AI, on-prem core

Platform on our VMs. AI calls go to Claude running in OUR OWN cloud tenancy over a private link — no public internet, no data retention.

+ Full capability + data control — routing already BUILT (one setting)

- Requires cloud tenancy sign-off

OPTION C

Public AI API

Platform on-prem but AI calls to the public API.

+ Simplest

- Fails our requirement — photos would traverse a public API

Recommendation: Option B — the software for BOTH A and B is already built; what remains is infrastructure sign-off.

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ARCHITECTURE

Target architecture — everything inside our walls

COMPANY INTERNAL NETWORK

Engineers (browser) + CAPEE
single sign-on (Azure AD)

Corporate API Gateway
authentication · rate limits · audit logs

CostVision server (frontend + backend on our VM)
geometry engine (CAD, in-memory) · 18 cost engines · learning loop · report generation

Corporate database (PostgreSQL)
rates · knowledge base · encrypted at rest

Internal AI gateway
ONLY allowed exit · inspected · logged

Our cloud tenancy
Claude on AWS Bedrock / Google Vertex
private link · no public internet · zero data retention

Zero-trust rules

- Every request authenticated (Azure AD)
- Deny-all egress except the AI gateway
- Role-based access (engineer / lead / admin / CAPEE service)
- CAD pulled from PLM vault, never stored

CAD DATA PROTECTION

How CAD data is protected

CAD model opened & measured	Inside our backend (OCCT engine)	✓ **Already local — verified**
Feature detection (holes, threads, walls)	Inside our backend	✓ **Already local — verified**
BOM extraction from files	Inside our backend (local parsers)	✓ **Already local — verified**
Cost calculation & learning	Inside our backend + our database	✓ **Already local — verified**
Photo analysis & AI narrative	Private AI endpoint (Option B)	→ **Routing BUILT — set one value**
CAD file storage	Nowhere — processed in memory only	✓ **Files never written to disk**

Plus: logs never contain CAD payloads; metadata goes to the corporate SIEM; and the "air-gapped" switch is now BUILT & tested — security can PROVE zero egress in a witnessed firewall test.

INTEGRATION

CAPEE + CostVision — better together

CAPEE (stays the front door)

- Costing workflow, approvals, reporting
- System of record — unchanged for users
- Calls CostVision services behind the scenes

requests →

← **AI answers**

CostVision (the engine behind it)

- 18 physics cost engines + CAD reading
- AI memory: similar parts, self-calibration
- Autonomous findings for sourcing

What flows over the connection (internal APIs that already exist)

Cost a part → full should-cost with breakdown

CAD file → geometry + auto-filled inputs

New part → similar past parts + suggestions

PO price → feeds the learning loop automatically

Dashboard ← autonomous findings (£/yr)

One shared → rate library & knowledge database

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SCOPE OF WORK

What each side needs to change

CostVision — 6 items (2 DONE)

✓ **Private AI routing —** DONE — one setting routes every AI call to our endpoint

✓ **"Air-gapped" switch —** DONE — provably disables all external calls; tested live

3. **Azure AD sign-on —** replace local logins with company SSO

4. **PostgreSQL —** move from file database to the corporate DB estate

5. **Cost API wrapper —** one clean endpoint per commodity for CAPEE

6. **Audit hardening —** admin audit table + logs to SIEM

CAPEE — small, additive changes

Backend · **Small**

HTTP client to call CostVision + token handling; hook PO prices into the learning API

User interface · **Medium**

an "AI insights" panel on the costing screen; a CAD-upload button; a findings widget

Data · **Small–Medium**

adopt the shared rate library; one-off import of historical quotes to seed the memory

Items 1–2 are built & tested. Remaining critical path: ~3–5 engineering weeks. No architectural surgery on either side.

COMPLIANCE

Security & compliance — how we tick the boxes

Encryption

TLS everywhere in transit; encrypted database at rest; CAD never stored at all

Access control

Azure AD single sign-on; roles mapped to AD groups; CAPEE uses a least-privilege service account

Data-loss prevention

One AI exit door, inspected and logged; firewall denies everything else; witnessed air-gap test (switch built)

Audit trails

Every rate change, knowledge write and admin action attributable to a user; logs to corporate SIEM

Standards mapping: ISO 27001 — covered by the controls above, add to ISMS scope · SOC 2 — inherited from the cloud tenancy (Bedrock/Vertex are certified) · GDPR — minimal personal data, zero-retention AI, EU region · ISO/SAE 21434 — engineering IT tool, out of vehicle scope; supply-chain review (SBOM) in CI.

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PLAN

Rollout — six phases, value from week eight

- **1 · Security assessment** 2–3 wks
- **2 · Architecture design** 3–4 wks
- **3 · Pilot (dummy CAD)** 2 wks
- **4 · CAPEE pilot** 4–6 wks
- **5 · Enterprise rollout** 4 wks
- **6 · Monitor & govern** ongoing

Phase 3 exit gate	Security-witnessed test: firewall blocks all egress, full test suite passes, zero unexpected traffic.
Phase 4 exit gate	A cost engineer completes a real costing from inside CAPEE with AI insights; accuracy dashboard live.
First value	Week ~8: pilot users get similar-part suggestions; sourcing sees the first autonomous findings.

After go-live we track: accuracy vs PO prices · costings via CAPEE · £ findings actioned · adoption. Approval today → pilot value by ~September 2026.

HONEST VIEW

Risks — and how we manage them

Cloud tenancy not approved

Likelihood: Medium

Mitigation: Fall back to Option A (air-gapped) — deterministic engines keep full value today; add a self-hosted vision model later if needed.

Learning depends on logged quotes

Likelihood: Medium

Mitigation: The CAPEE PO-price hook automates it — quotes flow in without anyone changing habits. Plus a one-off historical import to start smart.

Single-team knowledge of the platform

Likelihood: Low-Med

Mitigation: 1,005 automated tests, written architecture docs, and a named CAPEE-side maintainer trained during Phase 4.

Adoption ("another tool")

Likelihood: Low

Mitigation: Users stay in CAPEE — CostVision works behind the scenes. Nothing new to learn except better answers appearing.

DECISION

Feasibility verdict — and the ask

Secure deployment, CAD fully internal	**FEASIBLE — CAD never leaves; AI routing + air-gap switch already BUILT**
CAPEE + CostVision integration	**FEASIBLE — API-first design; CAPEE stays the front door**
Changes required	**Head start delivered: 2 of 6 items built — ~3-5 engineering weeks remain**
Long-term governance	**Rate-library board · monthly findings review · quarterly access & egress audit**

The ask today: approve Phase 1 (IT-Security assessment) and Phase 2 (architecture design) — 5–7 weeks, existing teams. The full written plan and security checklist are ready for the security review.

BACKUP · MARKET LANDSCAPE

Has anyone done this? Yes — the market is real

aPriori *Market leader (US)*	CAD-to-cost with physics-based models and regional cost databases. Used by Fortune-500 OEMs and Tier-1 suppliers — companies our size already pay for this category. *Source: apriori.com*
Siemens PCM *PLM giant (Teamcenter)*	Bottom-up should-costing with process models and supplier collaboration, embedded in the Siemens PLM suite. *Source: siemens.com/teamcenter*
Tset *AI challenger (Austria, 2018)*	Automotive-focused costing from 3D models and BOMs, 50+ calculation modules plus CO_2. Being acquired by A2MAC1 (June 2026) to build "AI-enabled costing intelligence". *Source: tset.com · globenewswire.com (18 Jun 2026)*
Boothroyd DFMA *The classic (Dewhurst)*	Design-for-manufacture and concurrent costing — the methodology textbooks are built on, used for decades. *Source: dfma.com*
New AI entrants *Start-ups*	Razorlabs Cost Advisor, Emithran and others: AI quotes from CAD files, calibrated on live machining data. *Source: emithran.com*

Takeaway: the should-cost category is proven — large OEMs already pay for it. The question is not IF it works, but why OURS fits us better: security, learning on our own data, and CAPEE integration (next slide).

BACKUP · DIFFERENTIATION

Why ours — what no vendor offers today

PCB photo → costed BOM

Photograph a circuit board, get a costed bill of materials with live component pricing. Design tools generate BOMs; no mainstream costing suite costs a board from photos.

Self-learning on OUR data

Calibrates itself on our real PO prices (error 10.9% → 0.3% after 3 quotes in testing) and raises findings autonomously. Incumbents are still marketing toward this.

Runs inside our walls

Commercial tools are cloud SaaS — our CAD would live in their cloud. CostVision is on-prem with private AI routing and a provable air-gapped switch, both already built.

One platform, no module licences

18 commodity engines + automotive software costing + carbon + RFQ generation in one codebase. Vendors sell comparable breadth as separately licensed modules.

Honest caveat: vendors like aPriori have decades of curated global cost data behind their numbers. Our answer is the learning loop: every quote CAPEE logs turns OUR history into a moat no vendor can buy — and our rates stay confidential.

Positioning: not "better than aPriori" — an internal, secure, self-learning costing engine behind CAPEE that gets smarter on our own data.

Sources: apriori.com · siemens.com/teamcenter · tset.com · globenewswire.com (A2MAC1-Tset, 18 Jun 2026) · dfma.com · emithran.com · flux.ai · circuitmind.io

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BACKUP · COMPANY COST DATABASE

Your rates, not vendor rates — already built

Rate Library — Edit & Version

🏭 **Company Rate Library** Company rates active company file uploaded

⬇ Download Excel template | ⬆ Upload company rates | Use built-in defaults | ↺ Reset to built-in

Materials

ID	GRADE	GBP/KG	SCRAP GBP/KG	REGION	CONF.
mat-al6061	6061-T6	3.62	0.55	UK	Medium
mat-steel1045	1045	0.95	0.22	UK	Medium
mat-ss316l	316L	3.82	0.85	UK	Medium
mat-steel4140	4140	1.21	0.22	UK	Medium
mat-ti6al4v	Ti-6Al-4V	47.5	16	UK	Low
mat-virtual	Virtual / Pass-through	1	0	UK	Medium
mat-en8	EN8 / 080M40 (Medium Carbon)	0.98	0.22	UK	Medium
mat-ss304-bar	304 Stainless Bar	3.4	0.8	UK	Medium
mat-ss303	303 Free-Machining Stainless Bar	3.7	0.8	UK	Medium
mat-al6082-bar	6082-T6 Aluminium Bar	3.55	0.55	UK	Medium
mat-al2011	2011-T3 Free-Machining Al Bar	3.95	0.55	UK	Low
mat-brass-cz121	CZ121 / CW614N Free-Machining Brass	7	3	UK	Low
mat-bronze-pb1	PB1 Phosphor Bronze Bar	9.2	2.8	UK	Low

Live screenshot (July 2026): Rate Library screen after a company workbook upload — badge shows "Company rates active".

How admins load company data

1. Download the Excel template — six sheets: Materials · Machines · Labour · Energy · FX · Overhead.
2. Fill in our rates and upload — every row is validated; the file becomes the active library instantly.
3. Fine-tune any single cell in the tables — each change is logged with the user's name and a timestamp.

Proven live today: a full workbook uploaded and accepted — 328 materials, 155 machines, 42 labour, 11 energy, 9 FX and 23 overhead rows — then activated.

Also built in:

- 20 country rate sets, 8 labour categories each
- PCB country cost table — admin-editable
- Separate rate library for software costing
- One click back to built-in defaults, full audit trail

BACKUP · BUSINESS CASE

What it costs vs what it returns

What it costs

Engineering — ~3–5 weeks remaining on CostVision + small additive CAPEE changes — existing teams

Infrastructure — one VM + corporate PostgreSQL — standard IT estate our teams already run

AI usage — pay-per-use in our own cloud tenancy — no per-seat fees, off in air-gap mode

Licence spend — £0 — built in-house, nothing to buy or renew

What it returns

Savings found — £512k/yr of pricing issues surfaced autonomously in the live demo (indicative)

Licence avoided — commercial should-cost suites run six figures per year, recurring

Speed — first defensible should-cost in minutes, not days — every quote challengeable

Leverage — bottom-up numbers on OUR rates that suppliers cannot wave away

OUR IP: built in-house — the code, the rate library and the knowledge base are entirely our intellectual property. No vendor owns any part of it.

Honest numbers: the £512k/yr figure comes from demonstration data — we treat it as indicative and validate it against real programmes during the Phase 3–4 pilot before claiming it in any business case.

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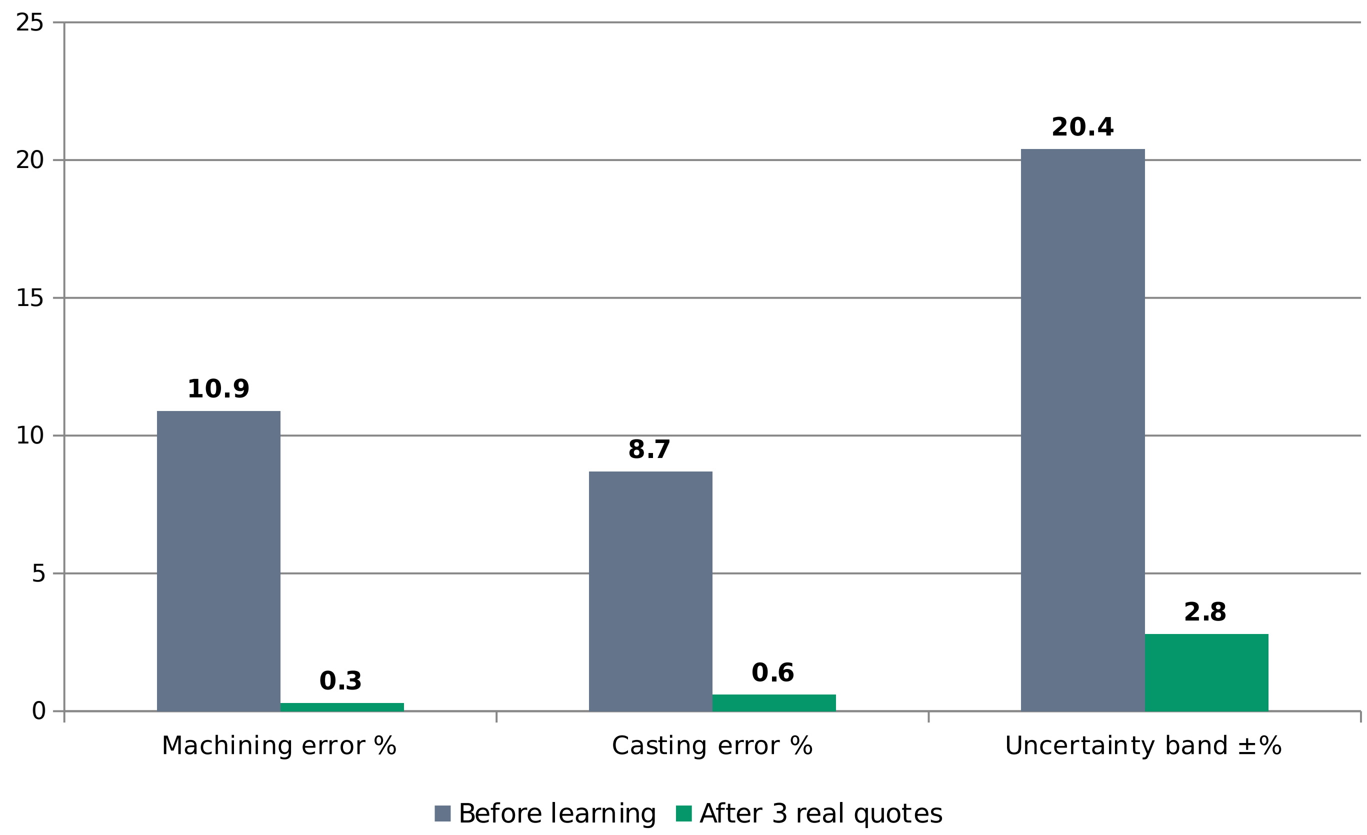
BACKUP · EVIDENCE

Does it actually work? The measured results



Live test runs, July 2026 — after the engine learned from just 3 real supplier quotes per commodity.

99% part-match accuracy — the AI memory finds the right similar past part

1,005 automated tests protect every engine and feature on each change

3 quotes is all the calibration needs before accuracy lands under 1%

£512k/yr findings raised by the autonomous agent in the live demo

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BACKUP · STRAIGHT ANSWERS

Questions you may be asking

Why not just buy aPriori?	A SaaS suite means our CAD and rates live in a vendor cloud, it cannot learn from our quote history, and it costs six figures every year. CostVision is on-prem, self-learning, and our IP.
Is our data training a public AI?	No. AI calls run zero-retention in our own cloud tenancy (Option B) — and in air-gapped mode there are provably no external calls at all. Nothing is ever used to train public models.
What if the key developer leaves?	1,005 automated tests define how everything must behave, the architecture is documented in writing, and Phase 4 trains a named CAPEE-side maintainer.
What does it cost to run?	One VM, a standard corporate database, and pay-per-use AI in our tenancy. No licences, no per-seat fees.
How do we know the numbers are right?	Physics-based cost build-ups on OUR uploaded rates, calibrated against OUR real PO prices — and the accuracy dashboard tracks the error openly, part by part.

BACKUP · FEATURE DEEP-DIVE

Photo to costed BOM — how it works, step by step

1 · Upload photos

Up to 5 photos of the board (+ an optional BOM or pick-and-place file). Photos are auto-downsized in the browser; nothing is stored.

2 · Board classification

AI identifies what the board is — domain (automotive, industrial, consumer) and safety level (ASIL) — so the right cost assumptions apply.

3 · OCR text pass

Every printed marking is read: chip part numbers, connector types, silkscreen references — hard evidence, not guesswork.

4 · Full vision BOM extraction

The AI locates and identifies every visible component — type, package, value, manufacturer part number (~20 seconds).

5 · Deterministic clean-up & costing

Code (not AI) normalises types, parses values, scores per-line confidence, grounds prices against the distributor catalogue, then adds board fabrication + SMT assembly + test on country rates.

6 · Same photos, same answer

The result is cached by a digital fingerprint (SHA-256) — re-running identical photos returns the identical BOM. Export PDF/Excel, save to library, feed the learning loop.

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BACKUP · FEATURE DEEP-DIVE

Photo to costed BOM — technology, accuracy, speed

Technology

- Claude vision model — via our controlled endpoint (Option B)
- 5-stage pipeline: classify → OCR → extract → cost → cache
- Deterministic TypeScript post-processor for types, values, MPNs
- SHA-256 result cache — repeatable, auditable output
- Optional live pricing (Nexar/RS/Farnell) — part numbers only, opt-in

Why it is accurate

- OCR evidence + vision cross-check — read, not guessed
- Code normalises every line; implausible part numbers rejected
- Per-line confidence score — low lines flagged for engineer review
- Prices grounded against the distributor catalogue, shown as ranges
- Learning loop calibrates against real quotes over time

Speed

- Main AI pass: ~20 seconds
- Full costed BOM: typically under a minute
- Repeat of the same board: instant (cache)
- Manual alternative: hours to days per board
- Works from photos alone — no CAD, no drawings needed

Honest limits: a vision BOM is a strong first pass, not gospel — hidden or underside parts need extra photos, and low-confidence lines are deliberately flagged for a human check rather than silently guessed.

BACKUP · FEATURE DEEP-DIVE

CAD file to cost — how it works, step by step

1 · Upload a CAD model
STEP, IGES or STL from any CAD system — as the standalone CAD-to-Cost flow or inside any of the 13 commodity calculators.

2 · Opened in memory, inside our server
The OCCT geometry kernel (the same engine class commercial CAD tools are built on) reads the model in memory. The file is never written to disk and never leaves the server.

3 · The model is measured
Exact volume, surface area, bounding box, mean wall thickness, hole count, free-form faces — measured from the geometry, not typed in from a drawing.

4 · Inputs auto-filled
Mass from volume × material density, stock/billet size, and suggested process parameters for the chosen commodity — the engineer reviews and adjusts.

5 · Physics engines cost it
Material, cycle or machining time, labour, tooling amortisation, energy and overheads — computed bottom-up on OUR uploaded rates for the chosen country.

6 · Defensible result
Full cost breakdown + DFM warnings (thin walls, tonnage limits) + an honest uncertainty band — and the learning loop tightens it as real quotes arrive.

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BACKUP · FEATURE DEEP-DIVE

CAD file to cost — technology, accuracy, speed

Technology

- OCCT (Open CASCADE) geometry kernel — runs inside our backend
- Pure-TypeScript STL fast path — no external process at all
- 18 deterministic physics cost engines do the actual costing
- AI translates the measured geometry into process inputs — it never receives the CAD file itself
- Rate library + 20-country regional rates drive every figure

Why it is accurate

- Measured, not estimated — exact volume → exact material mass
- Physics build-ups traceable line by line (no black box)
- DFM sanity checks: wall thickness, clamp tonnage, press force
- Honest uncertainty band shown with every result
- Calibration on 3 real quotes took machining error 10.9% → 0.3%

Speed

- Geometry read and measured in seconds
- Costing computes instantly once inputs are filled
- Model-to-first-price: a few minutes, mostly review time
- Manual alternative: hours of take-off per part
- Same flow in 13 commodities — one skill to learn

Security, restated: the CAD file is processed in memory inside our backend and is never stored or transmitted. Only a short numeric geometry summary reaches the AI layer — and in air-gapped mode, nothing leaves at all.